

Formulas TFE-4146

SEMICONDUCTOR PHYSICS

Electron Momentum: $p = m^*v = \hbar \mathbf{k} = \frac{h}{\lambda}$ Planck: $E = h\nu = \hbar\omega$

Kinetic: $E = \frac{1}{2} m^* v^2 = \frac{1}{2} \frac{p^2}{m^*} = \frac{\hbar^2}{2m^*} \mathbf{k}^2$ (3-4) Effective mass: $m^* = \frac{\hbar^2}{d^2 E / d \mathbf{k}^2}$ (3-3)

Total electron energy = P.E. + K.E. = $E_c + E(\mathbf{k})$

Fermi-Dirac e^- distribution: $f(E) = \frac{1}{e^{(E-E_F)/kT} + 1} \cong e^{-(E-E_F)/kT}$ for $E \gg E_F$ (3-10)

Equilibrium: $n_0 = \int_{E_c}^{\infty} f(E) N(E) dE = N_c f(E_c) = N_c e^{-(E_c - E_F)/kT}$ (3-15)

$N_c = 2 \left(\frac{2\pi m_n^* kT}{h^2} \right)^{3/2}$ $N_v = 2 \left(\frac{2\pi m_p^* kT}{h^2} \right)^{3/2}$ (3-16), (3-20)

$p_0 = N_v [1 - f(E_v)] = N_v e^{-(E_F - E_v)/kT}$ (3-19)

$n_i = N_c e^{-(E_i - E_v)/kT}$, $p_i = N_v e^{-(E_i - E_v)/kT}$ (3-21)

$n_i = \sqrt{N_c N_v} e^{-E_i/2kT} = 2 \left(\frac{2\pi kT}{h^2} \right)^{3/2} (m_n^* m_p^*)^{3/4} e^{-E_i/2kT}$ (3-23), (3-26)

Equilibrium: $n_0 = n_i e^{(E_F - E_i)/kT}$ $p_0 = n_i e^{(E_i - E_F)/kT}$ (3-25) $n_0 p_0 = n_i^2$ (3-24)

Steady state: $n = N_c e^{-(E_c - F_n)/kT} = n_i e^{(F_n - E_i)/kT}$ $p = N_v e^{-(F_p - E_v)/kT} = n_i e^{(E_i - F_p)/kT}$ (4-15) $np = n_i^2 e^{(F_n - F_p)/kT}$ (5-38)

$\mathcal{E}(x) = -\frac{dV(x)}{dx} = \frac{1}{q} \frac{dE_i}{dx}$ (4-26)

Poisson: $\frac{d\mathcal{E}(x)}{dx} = -\frac{d^2 V(x)}{dx^2} = \frac{\rho(x)}{\epsilon} = \frac{q}{\epsilon} (p - n + N_d^+ - N_a^-)$ (5-14)

$\mu \equiv \frac{q\bar{t}}{m^*}$ (3-40a) Drift: $v_d \cong \frac{\mu \mathcal{E}}{1 + \mu \mathcal{E}/v_s} \begin{cases} = \mu \mathcal{E} \text{ (low fields, ohmic)} \\ = v_s \text{ (high fields, saturated vel.)} \end{cases}$ (Fig. 6-9)

Drift current density: $\frac{I_x}{A} = J_x = q(n\mu_n + p\mu_p)\mathcal{E}_x = \sigma \mathcal{E}_x$ (3-43)

$$J_n(x) = q\mu_n n(x)\mathcal{E}(x) + qD_n \frac{dn(x)}{dx}$$

Conduction Current: drift diffusion (4-23)

$$J_p(x) = q\mu_p p(x)\mathcal{E}(x) - qD_p \frac{dp(x)}{dx}$$

$$J_{\text{total}} = J_{\text{conduction}} + J_{\text{displacement}} = J_n + J_p + C \frac{dV}{dt}$$

$$\text{Continuity: } \frac{\partial p(x, t)}{\partial t} = \frac{\partial \delta p}{\partial t} = -\frac{1}{q} \frac{\partial J_p}{\partial x} - \frac{\delta p}{\tau_p} \quad \frac{\partial \delta n}{\partial t} = \frac{1}{q} \frac{\partial J_n}{\partial x} - \frac{\delta n}{\tau_n} \quad (4-31)$$

$$\text{For steady state diffusion: } \frac{d^2 \delta n}{dx^2} = \frac{\delta n}{D_n \tau_n} \equiv \frac{\delta n}{L_n^2} \quad \frac{d^2 \delta p}{dx^2} = \frac{\delta p}{L_p^2} \quad (4-34)$$

$$\text{Diffusion length: } L \equiv \sqrt{D\tau} \quad \text{Einstein relation: } \frac{D}{\mu} = \frac{kT}{q} \quad (4-29)$$

p-n JUNCTIONS

$$\text{Equilibrium: } V_0 = \frac{kT}{q} \ln \frac{p_p}{p_n} = \frac{kT}{q} \ln \frac{N_a}{n_i^2/N_d} = \frac{kT}{q} \ln \frac{N_a N_d}{n_i^2} \quad (5-8)$$

$$\frac{p_p}{p_n} = \frac{n_n}{n_p} = e^{qV_0/kT} \quad (5-10) \quad W = \left[\frac{2\epsilon(V_0 - V)}{q} \left(\frac{N_a + N_d}{N_a N_d} \right) \right]^{1/2} \quad (5-57)$$

$$\text{One-sided abrupt } p^+-n: \quad x_{n0} = \frac{WN_a}{N_a + N_d} \approx W \quad (5-23) \quad V_0 = \frac{qN_d W^2}{2\epsilon}$$

$$\Delta p_n = p(x_{n0}) - p_n = p_n(e^{qV/kT} - 1) \quad (5-29)$$

$$\delta p(x_n) = \Delta p_n e^{-x_n/L_p} = p_n(e^{qV/kT} - 1)e^{-x_n/L_p} \quad (5-31b)$$

$$\text{Ideal diode: } I = qA \left(\frac{D_p}{L_p} p_n + \frac{D_n}{L_n} n_p \right) (e^{qV/kT} - 1) = I_0 (e^{qV/kT} - 1) \quad (5-36)$$

$$\text{Non-ideal: } I = I_0' (e^{qV/nkT} - 1) \quad (5-74) \\ (n = 1 \text{ to } 2)$$

$$\text{With light: } I_{\text{op}} = qA g_{\text{op}} (L_p + L_n + W) \quad (8-1)$$

$$I_D = \frac{\bar{\mu}_n Z C_i}{L} \left\{ (V_G - V_{FB} - 2\phi_F - \frac{1}{2}V_D)V_D - \frac{2}{3} \frac{\sqrt{2\epsilon_s q N_a}}{C_i} [(V_D + 2\phi_F)^{3/2} - (2\phi_F)^{3/2}] \right\} \quad (6-50)$$

$$I_D \approx \frac{\bar{\mu}_n Z C_i}{L} [(V_G - V_T)V_D - \frac{1}{2}V_D^2] \quad (6-49)$$

$$\text{Saturation: } I_D(\text{sat.}) \approx \frac{1}{2} \bar{\mu}_n C_i \frac{Z}{L} (V_G - V_T)^2 = \frac{Z}{2L} \bar{\mu}_n C_i V_D^2(\text{sat.}) \quad (6-53)$$

$$g_m = \frac{\partial I_D}{\partial V_G} ; \quad g_m(\text{sat.}) = \frac{\partial I_D(\text{sat.})}{\partial V_G} \approx \frac{Z}{L} \bar{\mu}_n C_i (V_G - V_T) \quad (6-54)$$

$$\text{For short } L: \quad I_D \approx Z C_i (V_G - V_T) v_s \quad (6-60)$$

$$\text{Subthreshold slope: } S = \frac{dV_G}{d(\log I_D)} = \frac{kT}{q} \ln 10 \left[1 + \frac{C_d + C_{it}}{C_i} \right] \quad (6-66)$$

BJT-p-n-p

$$I_{Ep} = qA \frac{D_p}{L_p} \left(\Delta p_E \operatorname{ctnh} \frac{W_b}{L_p} - \Delta p_C \operatorname{csch} \frac{W_b}{L_p} \right) \quad (7-18) \quad \begin{aligned} \Delta p_E &= p_n (e^{qV_{EB}/kT} - 1) \\ \Delta p_C &= p_n (e^{qV_{CB}/kT} - 1) \end{aligned} \quad (7-8)$$

$$I_C = qA \frac{D_p}{L_p} \left(\Delta p_E \operatorname{csch} \frac{W_b}{L_p} - \Delta p_C \operatorname{ctnh} \frac{W_b}{L_p} \right)$$

$$I_B = qA \frac{D_p}{L_p} \left[(\Delta p_E + \Delta p_C) \tanh \frac{W_b}{2L_p} \right] \quad (7-19)$$

$$B = \frac{I_C}{I_{Ep}} = \frac{\operatorname{csch} W_b/L_p}{\operatorname{ctnh} W_b/L_p} = \operatorname{sech} \frac{W_b}{L_p} \approx 1 - \left(\frac{W_b^2}{2L_p^2} \right) \quad (7-26)$$

(Base transport factor)

$$\gamma = \frac{I_{Ep}}{I_{En} + I_{Ep}} = \left[1 + \frac{L_p^n n_n \mu_n^p}{L_n^p p_p \mu_p^n} \tanh \frac{W_b}{L_p^n} \right]^{-1} \approx \left[1 + \frac{W_b n_n \mu_n^p}{L_n^p p_p \mu_p^n} \right]^{-1} \quad (7-25)$$

(Emitter injection efficiency)

$$\frac{i_C}{i_E} = B\gamma \equiv \alpha \quad (7-3)$$

$$\frac{i_C}{i_B} = \frac{B\gamma}{1 - B\gamma} = \frac{\alpha}{1 - \alpha} \equiv \beta \quad (7-6)$$

$$\frac{i_C}{i_B} = \beta = \frac{\tau_p}{\tau_t} \quad (7-7)$$

(Common base gain)

(Common emitter gain)

(For $\gamma = 1$)

Capacitance: $C = \left| \frac{dQ}{dV} \right|$ (5-55)

Junction Depletion: $C_j = \epsilon A \left[\frac{q}{2\epsilon(V_0 - V)} \frac{N_d N_a}{N_d + N_a} \right]^{1/2} = \frac{\epsilon A}{W}$ (5-62)

Stored charge
exp. hole dist.: $Q_p = qA \int_0^\infty \delta p(x_n) dx_n = qA \Delta p_n \int_0^\infty e^{-x_n/L_p} dx_n = qAL_p \Delta p_n$ (5-39)

$I_p(x_n = 0) = \frac{Q_p}{\tau_p} = qA \frac{L_p}{\tau_p} \Delta p_n = qA \frac{D_p}{L_p} p_n (e^{qV/kT} - 1)$ (5-40)

$G_s = \frac{dI}{dV} = \frac{qAL_p p_n}{\tau_p} \frac{d}{dV} (e^{qV/kT}) = \frac{q}{kT} I$ (5-67c)

Long p⁺-n: $i(t) = \frac{Q_p(t)}{\tau_p} + \frac{dQ_p(t)}{dt}$ (5-47)

MOS-n CHANNEL

Oxide: $C_i = \frac{\epsilon_i}{d}$ Depletion: $C_d = \frac{\epsilon_s}{W}$ MOS: $C = \frac{C_i C_d}{C_i + C_d}$ (6-36)

Threshold: $V_T = \underbrace{\Phi_{ms} - \frac{Q_i}{C_i}}_{\text{Flat band}} - \frac{Q_d}{C_i} + 2\phi_F$ (6-38)

Inversion: $\phi_s(\text{inv.}) = 2\phi_F = 2 \frac{kT}{q} \ln \frac{N_a}{n_i}$ (6-15) $W = \left[\frac{2\epsilon_s \phi_s}{qN_a} \right]^{1/2}$ (6-30)

$Q_d = -qN_a W_m = -2(\epsilon_s q N_a \phi_F)^{1/2}$ (6-32) At V_{FB} : $C_{FB} = \frac{C_i C_{\text{debye}}}{C_i + C_{\text{debye}}}$

Debye screening length: $L_D = \sqrt{\frac{\epsilon_s kT}{q^2 p_0}}$ (6-25) $C_{\text{debye}} = \frac{\epsilon_s}{L_D}$ (6-40)

Substrate bias: $\Delta V_T = \frac{\sqrt{2\epsilon_s q N_a}}{C_i} (-V_B)^{1/2}$ (n channel) (6-63)